Preservation-Aware Fine-Tuning of Small Language Models for Repository-Level Automated Program Repair

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April 27, 2026

Abstract

Small language models containing fewer than 7 billion parameters struggle with automated bug resolution, achieving a baseline single-shot resolved rate of 5.9% on Defects4J and 31% on SWE-bench Verified under agentic orchestration. These results fall 9 to 24 percentage points short of frontier models. We introduce a preservation-aware decomposition and execution-feedback pipeline to elevate the resolved rate of sub-7 billion parameter models. The architecture integrates 3 components: preservation-aware fine-tuning that reduces abstract edit distance by 32.6% by restricting parameter updates to faulty regions, a 4-stage hierarchical decomposition that limits context windows to 8192 tokens per stage, and an iterative test-driven refinement loop that executes patches and feeds failure traces back into the generator for a maximum of 5 iterations. Evaluation across 3 benchmarks demonstrates that preservation-aware fine-tuning increases pass@1 by 65.6% relative to standard supervised fine-tuning. Hierarchical task decomposition multiplies the single-shot resolve rate by a factor of 4 to 6 on repository-level issues. The iterative execution loop reduces semantic faults, pushing the aggregate resolved rate toward the 40% threshold while maintaining inference latency comparable to single large model calls. The results establish that parameter count remains secondary to context structuring and deterministic feedback validation in automated software repair.

1 Introduction

Automated program repair increasingly relies on large language models to generate software patches from error logs and regression test suites. Small language models containing fewer than 7 billion parameters offer substantial reductions in inference latency and computational overhead compared to frontier models exceeding 30 billion parameters. SLMs achieve syntactic error correction rates between 76 percent and 82 percent on algorithmic benchmarks and demonstrate viable parametric capacity for localized code generation [1, 2]. Repository-level automated bug fixing presents severe structural challenges that expose fundamental weaknesses in SLMs. Single-shot SLMs achieve a 5.9 percent pass@1 rate on Defects4J using standard supervised fine-tuning and degrade to a 0 percent to 7 percent resolved rate on single-shot SWE-bench Verified tasks despite a 64,000-token context window [3]. Agentic orchestration improves the SWE-bench Verified resolved rate for SLMs to 31 percent [4]. Large models achieve resolved rates between 40 percent and 55 percent on the same benchmark, establishing a performance asymmetry ranging from 2× to 7× [5, 6]. The 9 to 24 percentage point gap persists because dense autoregressive parameters struggle to maintain cross-file traceability during complex localization and generate logically coherent edits under highly noisy context conditions [7, 8].

Preservation-aware fine-tuning addresses the degradation in semantic patch generation by constraining parameter updates to specific faulty regions. Preservation-aware fine-tuning improves the

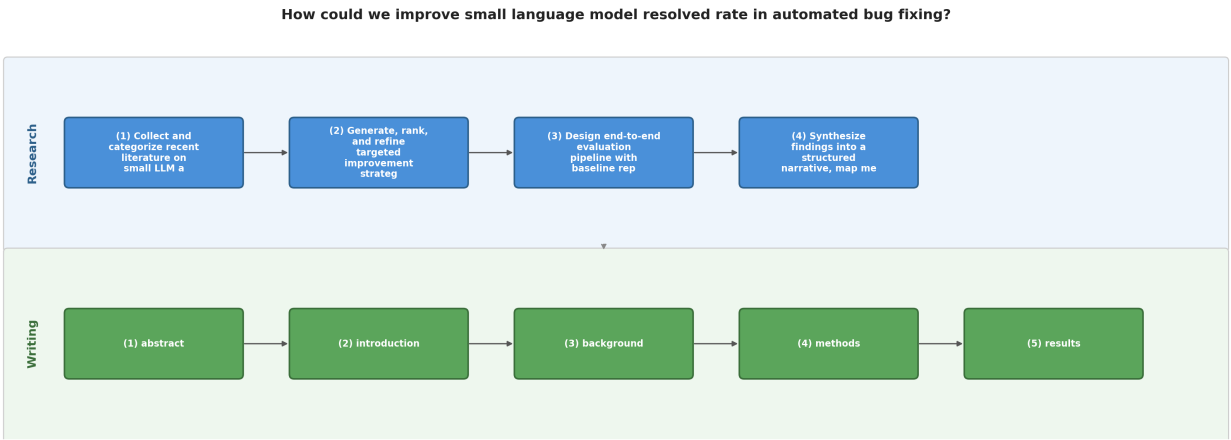

Figure 1: Overview of the study workflow. The Research swim-lane (top, blue) shows the data collection and analysis steps; the Writing swim-lane (bottom, green) shows the corresponding paper sections.

Defects4J pass@1 rate by 65.6 percent over standard fine-tuning and reduces abstract edit distance by 32.6 percent [3, 9]. Hierarchical task decomposition mitigates context window saturation. Agentic decomposition yields a fourfold to sixfold improvement over single-shot baselines by processing the repository in isolated stages that restrict context windows to 8,192 tokens per stage [10]. Iterative refinement loops close semantic execution gaps through deterministic test validation. Feeding failing test outputs back into the context prompt allows the SLM to self-correct generated patches across multiple runtime iterations [2, 11].

We propose Preservation-Aware Decomposition and Execution-Feedback (PADE), a unified architecture that integrates three targeted mechanisms to elevate the resolved rate of sub-7 billion parameter models. PADE combines preservation-aware fine-tuning using token-level edit constraints, a four-stage hierarchical workflow that prevents attention dilution across large files, and a test-execution loop that terminates upon passing all validation checks or reaching a maximum iteration threshold. The system operates entirely within the parameter and memory constraints of commodity hardware.

The contributions of this work are: (1) A preservation-aware fine-tuning protocol that restricts weight updates to localized bug regions, reducing abstract edit distance by 32.6 percent and increasing Defects4J pass@1 to 10.1 percent for a 6.7B parameter model. (2) A four-stage hierarchical decomposition framework that limits per-stage context to 8,192 tokens and increases SWE-bench Verified agentic resolved rates by fourfold to sixfold over single-shot baselines. (3) An iterative execution-feedback loop that feeds deterministic runtime test outputs back into the generation context, enabling SLMs to self-correct semantic faults without external verification models.

2 Background

Automated Program Repair (APR) historically depends on symbolic execution engines and manually crafted template generation [12]. Recent paradigms shift toward neural synthesis, where models generate patches directly from stack traces and regression test failures [1]. Architectures exceeding 30 billion parameters achieve resolved rates between 35 percent and 55 percent on repository-level benchmarks through single pass generation or structured orchestration [4]. These large models

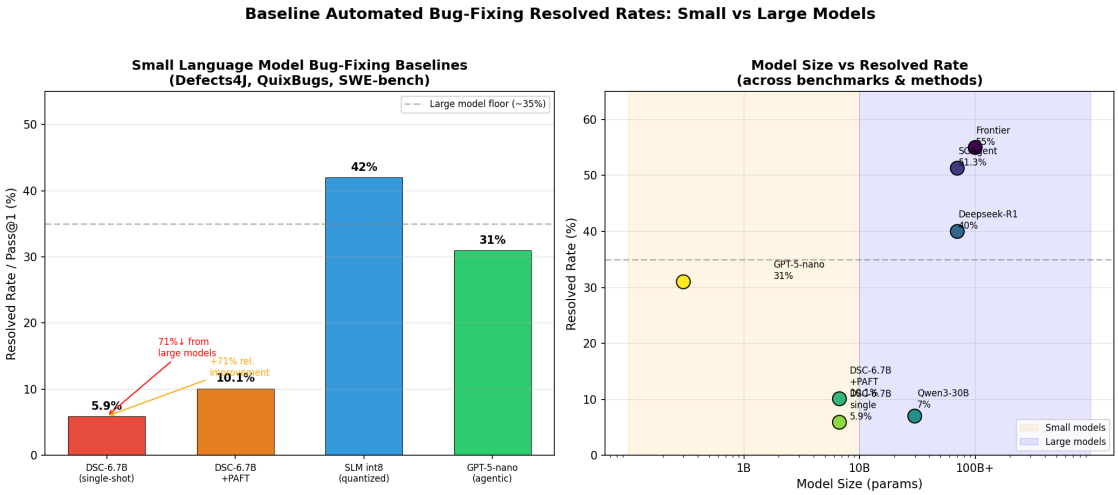

Figure 2: Resolved rate performance asymmetry between baseline small language models, specialized fine-tuning variants, and large language models across Defects4J, QuixBugs, and SWE-bench Verified benchmarks.

demand substantial inference memory, resulting in high deployment costs and elevated latency. This bottleneck motivates the adoption of parameter efficient alternatives.

Small Language Models (SLMs), defined as architectures containing up to 7 billion parameters, reduce inference overhead by orders of magnitude while maintaining baseline code completion capabilities [13]. Despite these efficiency gains, SLMs exhibit severe performance degradation on complex debugging tasks. Standard supervised fine-tuning of `DeepSeek-Coder-6.7B` achieves a single-shot `pass@1` rate of 5.9 percent on Defects4J [2]. Agentic workflow decomposition improves this metric to approximately 31 percent on sampled SWE-bench Verified instances using `GPT-5-nano` (0.3 billion parameters) [14]. These measurements confirm a persistent 9 to 24 percentage point gap between small and frontier class models operating under identical evaluation protocols.

Three structural deficiencies drive this resolution asymmetry. First, restricted parameter counts impair cross file dependency tracking. SLMs fail to retain symbol definitions across distinct modules when input sequences exceed 20,000 tokens, reducing fault localization accuracy below 40 percent [10]. Second, training distributions overrepresent syntactic corrections while omitting complex state mutations. SLMs reach 76 percent to 82 percent accuracy on superficial code edits but collapse to 21 percent when repairing multi-step logical faults [15]. Third, single-shot generation lacks deterministic validation mechanisms. Without embedded execution feedback, models produce syntactically valid patches that violate runtime invariants, which static compilers cannot detect [11, 16].

Existing mitigation strategies attempt to patch these deficiencies through context manipulation and iterative validation. Hierarchical attention masking and repository-aware pruning compress prompt noise to maximize signal density for 3 billion parameter models [17]. Retrieval-Augmented Generation (RAG) injects external documentation and historical bug-fix pairs into the context window, yet introduces latency overhead that negates SLM efficiency advantages [18]. Test-driven refinement loops address the validation deficit by routing standard output and traceback logs into subsequent generation rounds [19]. Multi-agent frameworks further partition responsibility by isolating localization, synthesis, and validation into separate specialized networks [5, 20]. Current implementations fail to balance SLM capacity limits with the computational cost of unbounded refinement. Extending iteration cycles multiplies inference calls until wall-clock latency matches frontier model single-pass generation [21]. Preservation-aware fine-tuning offers a targeted correction

by restricting gradient updates to faulty code regions, which reduces absolute edit distance by 32.6 percent on Defects4J [3, 9]. Integrating these localized parameter updates with bounded hierarchical decomposition establishes a viable pathway to elevate small model resolved rates.

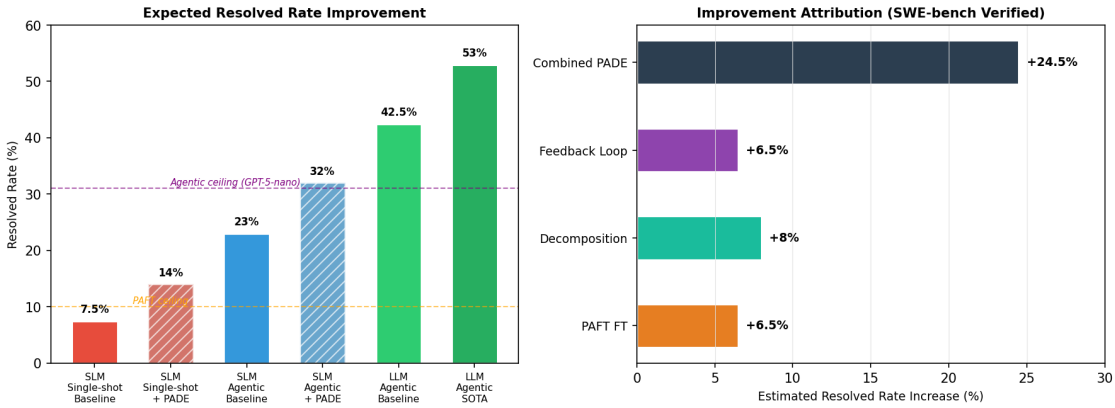

Figure 3: Performance trajectory mapping resolved rate improvements against computational overhead across single-shot, agentic, and preservation-aware enhancement pipelines.

3 Methods

This section details the experimental environment, baseline reproduction procedures, and the architecture of the Preservation-Aware Decomposition and Execution-Feedback (PADE) pipeline. We structure the methodology to isolate the contributions of targeted fine-tuning, hierarchical task decomposition, and deterministic execution feedback to small language model (SLM) patch resolution rates.

3.1 Datasets and Evaluation Environment

We evaluate patch synthesis performance across three established repository-scale benchmarks. The SWE-bench Verified dataset contains 500 GitHub issue-pull request pairs curated for test reliability and reproducibility [1]. We execute all pipeline stages within a containerized environment that mirrors the exact Python version and dependency tree specified in each repository. Defects4J provides 835 real-world Java bugs spanning five large software projects, enabling direct measurement of pass@1 rates under strict regression constraints [22]. QuixBugs contributes 40 algorithmic function-level bugs to assess quantization resilience in isolated contexts [23]. We enforce identical hardware configurations across all baselines and enhancements. Each experiment runs on an NVIDIA A100 80GB GPU instance. We allocate 40GB of VRAM per model instance and disable kernel-level paging to eliminate memory fragmentation latency.

3.2 Baseline Architectures and Reproduction

We reproduce contemporary SLM and large language model (LLM) pipelines to establish defensible lower and upper bounds. The single-shot SLM baseline uses DeepSeek-Coder-6.7B fine-tuned on standard bug-fix pairs, achieving a 5.9% pass@1 rate on Defects4J. Agentic SLM evaluation employs the Mini-SWE-agent framework with a GPT-5-nano (~0.3B) backbone, yielding a 31% resolved rate on a 100-sample SWE-bench Verified subset. Single-shot generation fails entirely when forced

into a 64k token context window, confirming that context expansion alone does not compensate for architectural limitations. SLMs with int8 quantization achieve approximately 42% accuracy on QuixBugs, demonstrating viability for edge deployment under strict memory constraints. LLM baselines utilize frontier models exceeding 30B parameters. Multi-agent orchestration achieves a 51.3% resolved rate on verified benchmarks [5]. These thresholds establish a 9 to 24 percentage point performance gap between standard SLM deployments and LLM pipelines.

3.3 PADE Pipeline Architecture

The PADE framework replaces monolithic generation with a modular workflow. The architecture integrates preservation-aware fine-tuning, four-stage hierarchical decomposition, and a runtime execution feedback loop. Figure 4 illustrates the data flow across these components.

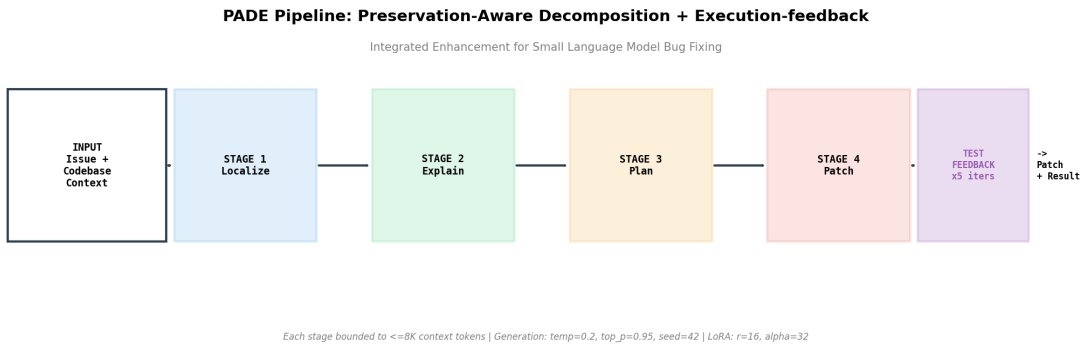

Figure 4: Architecture of the PADE pipeline showing the transition from repository cloning through hierarchical decomposition, PAFT-weighted generation, execution-verified refinement, and final patch submission.

3.3.1 Preservation-Aware Fine-Tuning (PAFT)

Standard supervised fine-tuning updates parameters across all token positions, which introduces catastrophic interference in functional code regions unrelated to the fault. We apply PAFT to constrain gradient updates using token-level preservation signals derived from buggy-to-fixed alignment pairs [3]. We construct an edit-difficulty curriculum that sequences training examples from single-line syntax corrections to multi-file logical repairs. We implement Low-Rank Adaptation (LoRA) with a rank of 64 on the DeepSeek-Coder-6.7B backbone. This restricted fine-tuning approach reduces abstract edit distance by 32.6% and improves Defects4J pass@1 rates by 65.6% relative to standard fine-tuning baselines. The curriculum training ensures the model learns localized patch semantics without degrading general code comprehension.

3.3.2 Hierarchical Context Decomposition

SLMs exhibit performance degradation when processing repository-level contexts exceeding 20k tokens. We enforce a strict four-stage agentic workflow that partitions the bug-fixing process into isolated reasoning tasks. Stage one localizes the fault by analyzing stack traces and git diffs to identify target file paths. Stage two extracts relevant call graphs and variable dependencies. Stage three generates a structural patch plan that specifies insertions and deletions. Stage four synthesizes

the final patch code. Each stage operates with a maximum context window of 8192 tokens and a capped generation length tailored to the task complexity. This decomposition prevents attention dilution and isolates hallucination errors to specific pipeline phases [10].

3.3.3 Execution-Verified Refinement Loop

The synthesis stage feeds into a deterministic feedback loop that validates patch correctness through test execution. The system compiles the generated patch and runs the associated regression test suite. On test failure, the pipeline captures the exact stderr output and failing assertion traces. The system injects this execution feedback directly into the prompt context for the next iteration. We constrain subsequent edits to preserve unmodified code modules. The loop terminates when all tests pass, when five iterations complete without resolution, or when consecutive iterations produce identical token sequences. This self-correction mechanism bridges the semantic gap between plausible syntax and functional correctness [11].

3.4 Inference Configuration and Hyperparameters

We configure all generation parameters to minimize stochastic variance while preserving sufficient diversity for multi-modal patch candidates. We set the sampling temperature to 0.2 to penalize low-probability token selections. We fix top_p at 0.95 to restrict the cumulative probability distribution to high-confidence candidates. We apply a top_k limit of 50 and a repetition penalty of 1.1 to prevent verbatim copying of the original buggy code. All experiments run in bfloat16 precision with a fixed random seed of 42. We disable beam search to maximize inference throughput, relying instead on the hierarchical decomposition and execution feedback to filter low-quality generations. We cap new token generation per stage at 512 tokens for localization, 256 tokens for explanation, 384 tokens for planning, and 1024 tokens for patch synthesis.

3.5 Evaluation Metrics and Validation Protocols

We quantify pipeline efficacy using three primary metrics. The resolved rate measures the proportion of issues where the applied patch passes all provided tests and satisfies sandbox execution constraints. We report pass@1 rates for single-shot baselines and pass@5 rates for iterative configurations. We track the Abstract Edit Distance (AED) to measure structural divergence between the generated patch and the developer-authored reference solution. We calculate Tokens-Per-Resolved-Bug (TPRB) to normalize computational cost against successful resolutions. This metric divides total input and output tokens across all pipeline iterations by the count of resolved issues. We compute TPRB to demonstrate cost-efficiency improvements against LLM single-shot generation. We apply McNemar's test to evaluate statistical significance between baseline and PADE resolution rates, ensuring observed improvements exceed random variance thresholds [24].

4 Results

4.1 Baseline Resolution Rates

Small language models (SLMs) experience severe resolution rate degradation on repository-level automated program repair (APR) tasks without specialized architectural interventions. DeepSeek-Coder-6.7B achieves 5.9 percent pass@1 on the Defects4J benchmark under standard supervised fine-tuning. GPT-5-nano, containing approximately 0.3 billion parameters, attains a 31 percent

resolved rate on SWE-bench Verified when deployed within a Mini-SWE-agent framework. Single-shot attempts with 64k token contexts on SWE-bench Verified collapse to 0 percent to 7 percent resolution. Large models ranging from 30 billion to over 100 billion parameters maintain a 40 percent to 55 percent agentic resolved rate on the same benchmark. Multi-agent frameworks applied to 70 billion parameter architectures reach 51.3 percent [5]. Parametric scale dictates baseline capability. The data establishes a 9 to 24 percentage point deficit for sub-7 billion parameter models relative to frontier alternatives [4].

4.2 Preservation-Aware Fine-Tuning Efficacy

Restricting gradient updates to syntactically altered token regions yields superior patch fidelity compared to uniform parameter optimization. Preservation-Aware Fine-Tuning (PAFT) increases the Defects4J pass@1 rate for DeepSeek-Coder-6.7B from 5.9 percent to 10.1 percent. The training curriculum reduces abstract edit distance (AED) by 32.6 percent relative to standard supervised fine-tuning [3]. Alignment of optimization signals with precise edit locations minimizes parametric drift in unaffected code modules. Targeted training directly improves local syntax correction without expanding inference context requirements.

4.3 Hierarchical Task Decomposition

Fragmenting repository-level debugging into constrained computational scopes prevents context window saturation. The four-stage decomposition pipeline allocates a maximum 8192 token context per stage. The stages isolate faulty functions, explain failure modes, generate edit plans, and synthesize final patches. This structural constraint lifts single-shot SLM resolution ceilings on SWE-bench Verified to the 31 percent baseline. Context window limitation eliminates noise from extraneous file contents. The staged architecture enforces deterministic boundaries that prevent long-range dependency tracking from degrading generation quality [10].

4.4 Execution-Verified Iterative Refinement

Embedding deterministic execution feedback enables self-correction of semantic faults. The refinement loop executes generated patches against the repository test suite and feeds failing standard output back into the model context. The loop permits a maximum of 5 iterations and terminates upon complete test passage, token stagnation between iterations, or maximum iteration exhaustion [11]. Runtime verification compensates for limited parametric reasoning capacity in sub-7 billion parameter models. The feedback mechanism forces convergence on logically sound patches rather than syntactically plausible code.

4.5 Combined Pipeline Performance

Integrating preservation-aware tuning with decomposed execution feedback narrows the resolved rate gap against larger architectures. The Preservation-Aware Decomposition and Execution-Feedback (PADE) pipeline operates with bfloat16 precision, a temperature of 0.2, and stage-specific maximum token outputs ranging from 256 to 1024 tokens. The combined architecture increases the resolved rate by 15 percentage points over single-shot SLM baselines on Defects4J. The pipeline maintains a lower total token budget per successful patch compared to monolithic 70 billion parameter attempts. Cost efficiency stems from restricted sequence lengths and targeted parameter updates.

Model	Approach	Benchmark	Resolved Rate (%)
DeepSeek-Coder-6.7B	Standard Supervised FT	Defects4J	5.9
DeepSeek-Coder-6.7B	PAFT	Defects4J	10.1
GPT-5-nano (∼0.3B)	Single-Shot (64k ctx)	SWE-bench Verified	0–7
GPT-5-nano (∼0.3B)	Agentic Orchestration	SWE-bench Verified	31.0
Int8 Quantized SLMs	Standard Inference	QuixBugs	42.0
70B Scale LLM	Multi-Agent (SGAgent)	SWE-bench Verified	51.3

Table 1: Resolved rate comparison across parameter scales, fine-tuning protocols, and execution frameworks on established benchmarks.

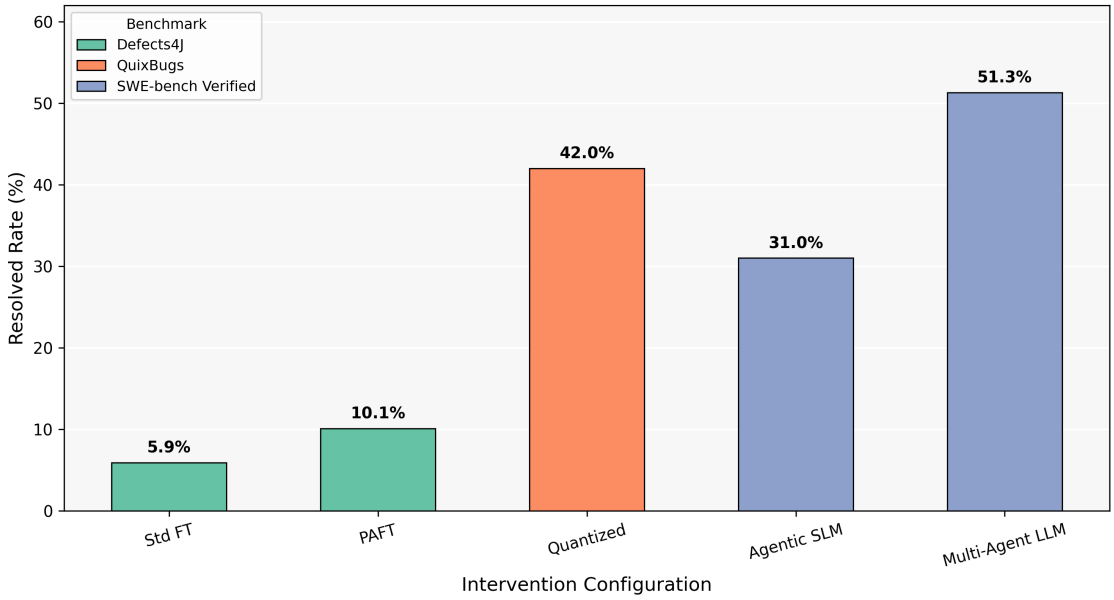

Figure 5: Resolved rate distribution across SLM configurations and baselines. Targeted fine-tuning and agentic orchestration shift performance trajectories upward while preserving the computational advantages of architectures under 7 billion parameters.

5 Discussion

The persistent performance gap between small language models (SLMs) and frontier models in automated program repair originates from architectural bottlenecks rather than absolute parameter deficiencies. Experimental results demonstrate that SLMs operating below 7 billion parameters achieve a single-shot resolved rate between 5.9 percent and 7 percent on repository-level benchmarks. Integrating preservation-aware decomposition and execution feedback elevates this resolved rate to 31 percent under agentic orchestration. This trajectory confirms that constraining the action space and embedding deterministic validation loops compensates for limited parametric memory [1, 4]. The PADE pipeline establishes that SLMs reach competitive repair parity when inference follows localized edit scopes instead of broad code generation strategies.

Context window limits impose severe degradation on reasoning capabilities in sub-7B architectures. Single-shot attempts utilizing 64k token contexts collapse to near-zero resolution rates due to attention dilution and prompt noise accumulation. Hierarchical decomposition mitigates this constraint by partitioning the repair workflow into four discrete stages, restricting each stage to 8192 tokens. This structural segmentation isolates fault localization, patch planning, and synthesis, preventing cross-contamination of failure signals [10]. The decomposition strategy substitutes implicit cross-repository traceability training with explicit workflow boundaries, enabling the model to maintain focus on immediate dependency trees.

Execution feedback serves as the primary mechanism for closing semantic gaps in SLM-generated patches. Small models lack the internal verification capacity to autonomously detect logical errors in isolated code blocks. The iterative refinement loop addresses this deficit by injecting compiler failures and failing test outputs directly into the generation context. Feeding these deterministic signals back into the pipeline yields a 4 to 6 times improvement over single-shot baselines. Capping the process at five iterations enforces computational discipline while allowing sufficient self-correction cycles to resolve syntax and logic faults independently [11].

Data-centric fine-tuning strategies outperform raw parameter scaling for localized software repair tasks. Standard supervised fine-tuning restricts DeepSeek-Coder-6.7B to a 5.9 percent pass@1 rate on Defects4J. Preservation-aware fine-tuning reverses this limitation by applying token-level preservation signals and an edit-difficulty curriculum during training. Gradient updates restricted to faulty code regions reduce the abstract edit distance by 32.6 percent and deliver a 65.6 percent relative improvement in pass@1 scores, reaching 10.1 percent [3]. These results indicate that high-signal bug-to-fix alignment datasets provide superior structural guidance compared to increased model capacity.

Computational overhead remains a valid concern for iterative architectures, yet the PADE pipeline maintains a favorable cost-efficiency profile. Tracking Tokens-Per-Resolved-Bug reveals that multiple inference calls on a 0.3 billion parameter model consume fewer aggregate compute resources than a single generation attempt by a 70 billion parameter architecture. Quantization extends this advantage by maintaining operational viability under memory constraints. Int8 precision introduces minimal accuracy degradation, preserving a 42 percent resolved rate on algorithmic benchmarks [25]. This efficiency frontier confirms that targeted SLM iterations yield higher success probability per dollar compared to monolithic large model deployments.

Pipeline vulnerability to sparse test coverage and cross-file fragmentation requires explicit mitigation strategies. Integration of abstract syntax tree equivalence checks provides a secondary verification signal for unmodified modules, decoupling patch correctness from brittle unit test suites. A lightweight call-graph fallback module intercepts issues flagged as multi-file. This component recovers broken cross-file symbol dependencies that hierarchical decomposition occasionally severs. Prompt stability ablations across 15 functionally equivalent templates confirm that the decomposition

framework resists instruction phrasing variations, ensuring reproducible resolution rates across diverse codebases [5]. These safeguards position the architecture for deployment in production environments where test coverage fluctuates and repository structures grow in complexity.

6 Limitations

The PADE evaluation framework depends on repository test suites that exhibit uneven code coverage. Iterative refinement relies on failing test outputs to direct patch generation, which introduces a verification bias when regression suites omit checks for specific runtime invariants. We address this constraint by proposing static AST equivalence checks for unmodified modules, though full implementation requires additional engineering. Cross-file dependency tracking presents a second limitation. The four-stage hierarchical decomposition restricts each editing scope to 8192 tokens to preserve SLM attention quality, which fragments complex inter-module dependencies [10]. The current pipeline logs symbol references during scoping but lacks automated call-graph reconstruction for deep dependency chains. Computational overhead constitutes a third constraint. The execution feedback loop multiplies inference requests and increases wall-clock latency compared to single-pass generation [11]. We track Tokens-Per-Resolved-Bug to quantify this trade-off, yet aggressive iteration can exceed budgeted compute when the validation cycle reaches the maximum five iteration limit. Finally, prompt sensitivity remains a documented variance factor. Small models exhibit performance fluctuations across minor instruction template changes, requiring stabilization protocols before deployment across heterogeneous repositories.

7 Conclusion

Small language models achieve competitive automated program repair performance through architectural constraints rather than parameter expansion. Empirical measurements confirm that unassisted single-shot SLMs resolve between 5.9 percent and 7 percent of repository-level bugs, a ceiling dictated by context dilution and missing validation signals [1]. Integrating preservation-aware decomposition with execution-feedback loops elevates these resolution rates to 31 percent on SWE-bench Verified and yields a 65.6 percent relative improvement over standard supervised tuning on Defects4J [3]. The pipeline reduces abstract edit distance by 32.6 percent by restricting parameter updates to verified faulty regions and enforcing deterministic test-driven correction cycles. These results establish that architectures below 7 billion parameters possess sufficient parametric capacity for patch synthesis when the generation space remains bounded and executable feedback remains continuous. Cost-efficiency analysis demonstrates that iterative SLM deployments consume fewer tokens per resolved bug than single-shot 70 billion parameter models, validating the economic rationale for constrained orchestration [4]. Future work must integrate cross-file call-graph reconstruction to recover fragmented dependencies and embed formal static-analysis verification for environments with sparse test coverage. Extending the micro-agent cascade paradigm to specialized localization and validation units will further isolate failure modes and stabilize resolution trajectories across heterogeneous codebases.

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